3.2 Terms and Co-efficients

We combine variables and constants using the mathematical operations addition and subtraction to construct **algebraic expressions.**

For example, the expression $6x + 1$ is obtained by adding two parts $6x$ and 1. Here $6x$ and 1 are known as **terms.** The term $6x$ is a variable and the term 1 is a constant, since it is not multiplied by a variable. Also we say that 6, x are the factors of the term $6x$.

Similarly, in the expression $3ab + 5c$, the terms are $3ab$ and $5c$. The factors of the term $3ab$ are 3, a and b. In the same way, the factors of the term $5c$ are 5 and c.

Now let us further try to understand the terms of an algebraic expression.

Observe the Fig.3.2

What is the perimeter of the given figure?

The perimeter, $'P' = x + 5 + 6 + y + 5 + 4 + z + w \ units,$

$= x + y + z + w + (5 + 6 + 5 + 4)$

$= x + y + z + w + 20,$

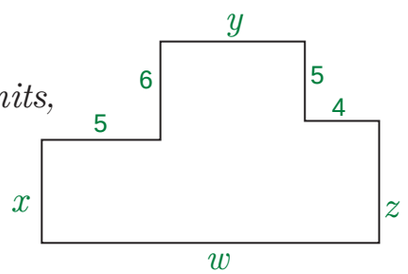


Fig.3.2

where x, y, z, w are variables and 20 is a constant.

Note that, in the above expression, 5 terms are combined by using addition.

Consider another example as $6x - 5y + 3$. To find the terms of the expression, we write $6x + (-5y) + 3$. Here the terms are $6x, (-5y)$ and 3. An expression may have one, two, three or more terms.

Think

Can we use the operations multiplication and division to combine terms?

Also, a term may be any one of the following:

i) a constant such as 8, −11, 7, −1,...

ii) a variable such as $x, a, p, y, ...$

iii) a product of two or more variables such as $xy, pq, abc, ...$

iv) a product of constant and a variable/variables such as $5x, -7pq, 3abc, ...$

Note

An algebraic expression can have one term, two terms or more than two terms. An expression with one term is called a **monomial**, two terms is called a **binomial** and three terms is called a **trinomial**. An expression with one or more terms is called a **polynomial.** For example, the expression $2x$ is a **monomial,** $2x + 3y$ is a **binomial,** and $2x + 3y + 4z$ is a **trinomial.** All the expressions given above are polynomials.

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